

Stochastic means: when a random pattern occurs that may be analysed statistically but may not be predicted accurately.

Stochastic process is to understand various tools and techniques available to model various stochastic scenarios that we face in life.

Ex:

If we want to go from one place to other, the travel time depends on

i) traffic ii) weather condition etc.

We want to compute average travel time.

For this we need various tools.

Example 1 :

Consider an experiment of throwing a die infinitely many times.

$$\mathcal{P} = \left\{ (5, 1, 4, 4, 5, \dots), \right. \\ (1, 1, 1, 1, \dots), \\ \left. (2, 1, \dots) \dots \right\}$$

Define $X_i: \mathcal{P} \rightarrow \mathbb{R}$ by

X_1 = outcome of first throw. $\mathcal{A} = \{1, 2, \dots, 6\}$

X_2 = ——— 2nd throw $\mathcal{A}_2 = \{1, 2, \dots, 6\}$

$\vdots$

i) For $n \in \mathbb{N}$, $X_n: \mathcal{P} \rightarrow \mathbb{R}$ by
 X_n is outcome of the n^{th} throw.
 $\mathcal{A}_n = \{1, 2, 3, 4, 5, 6\}$

$\{X_n \mid n \geq 1\} \rightarrow$ Family of random variables.

ii) $X_n: \mathcal{P} \rightarrow \mathbb{R}$ X_n is the number of 6's in n throws.

$\mathcal{A}_1$ for X_1 is $\mathcal{A}_1 = \{0, 1\}$

$\mathcal{A}_2$ for X_2 is $\mathcal{A}_2 = \{0, 1, 2\}$

$\mathcal{A}_n$ for X_n is $\mathcal{A}_n = \{0, 1, 2, \dots, n\}$

$\{X_n \mid n \geq 1\} \rightarrow$ Family of r.v.

Example 2:

Consider an experiment of no of telephone calls received at a telephone exchange.

$$\mathcal{P} = \{0, 1, 2, 3, \dots\}$$

Define $X(t) : \mathcal{P} \rightarrow \mathbb{R}$ by number of calls received in time interval $(0, t)$

$$X(1 \text{ sec}) \quad \mathcal{A} = \{0, 1, 2, \dots\}$$

$$X(1.5 \text{ sec}) \quad \mathcal{A} = \{0, 1, 2, \dots\}$$

$\{X(t) \mid t \geq 0\}$ is family of r.v.

Stochastic Process : A family of random variables which are function of say time are known as stochastic processes.

State space: The set of possible values of a single variable X_i of a stochastic process $\{X_n \mid n \geq 1\}$ is called state space.

Time parameter: The set of points at which the process is observed.

Types of stochastic processes:

i) Discrete time, discrete state space.

Example:

- Consider an experiment of throwing a die infinitely many times.

- X_n is outcome of the n^{th} throw.

$$\Omega_n = \{1, 2, 3, 4, 5, 6\}$$

$\{X_n \mid n \geq 1\}$ is an example.

ii) Continuous time, discrete state space

Example: $x(t) : \mathcal{P} \rightarrow \mathbb{R}$ by number of calls received in time interval $(0, t)$

$$\{x(t) \mid t \geq 0\}$$

iii) Continuous time, continuous state space

Ex: $x(t)$ be maximum temperature at a place in time interval $(0, t)$

$$\{x(t) \mid t \geq 0\}$$

iv) Discrete time, continuous state space:

Ex: X_n be temperature measured at 12 pm on n^{th} day.

$$\{X_n \mid n \in \mathbb{N}\}$$

Remark :

1. All the 4 processes can be represented as $\{X(t) \mid t \in T\}$.

• $T = \mathbb{N}$ or $T = (0, \infty)$

2. t - time, distance, length, number of trials.

Important question:

For family of r.v $\{X_n \mid n \geq 1\}$,
are X_i independent of each other or dependent?

Rarely they are independent:

Example:

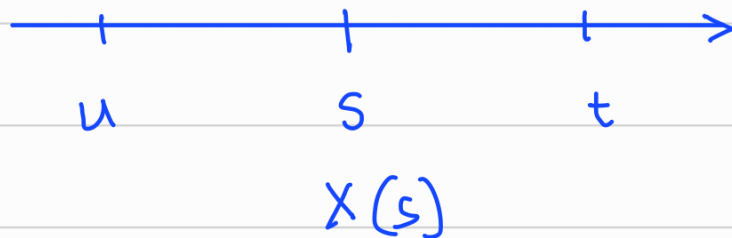
X_n is outcome of the n^{th} throw.

Most of the times they are dependent.

Based on the dependency, we have Markov process.

Markov process:

If $\{x(t) \mid t \in T\}$ is a stochastic process s.t., given the value $x(s)$, the values $x(t)$, $t \geq s$ do not depend on the values of $x(u)$, where $u < s$. Then such a process is called "Markov process".



In other words, knowing the present makes the past irrelevant for predicting the future.

Ex: Consider experiment of tossing a fair coin repeatedly.

For each toss outcomes is $\{H, T\}$.

$\downarrow \quad \downarrow$
probability $\rightarrow P \quad q$

$$p + q = 1.$$

• Let us denote Head by 1
Tail by 0.

Define X_n = result of n^{th} toss.

Question :

$$\begin{aligned} P_{X_n}(X_n = 1) &= p \\ P(X_n = 0) &= q \end{aligned}$$

Here $X_1, X_2, \dots$ are independent variables. This is Markov process.

Consider another r.v. given by

$$S_n = X_1 + X_2 + \dots + X_n.$$

Note S_n gives accumulated no. of heads in the first n -trials & its possible values are $\{0, 1, 2, \dots, n\}$.

$$\begin{aligned}\text{Note that } S_{n+1} &= (X_1 + \dots + X_n) + X_{n+1} \\ &= S_n + X_{n+1}.\end{aligned}$$

$$\begin{aligned}\text{Suppose } S_n = j = \sum_{i=1}^n X_i \quad j = (0, 1, 2, \dots, n) \\ S_{n+1} = X_1 + \dots + X_n + X_{n+1} =\end{aligned}$$

Then, S_{n+1} can have two values only

$$\underline{S_{n+1} = j} \quad \text{iff} \quad \underline{X_{n+1} = 0} \quad \text{and}$$

$$S_{n+1} = S_n + X_{n+1} = \underline{j+1} \quad \text{iff} \quad \underline{X_{n+1} = 1}.$$

$$\begin{aligned}\therefore P(S_{n+1} = j \mid S_n = j) &= P(X_{n+1} = 0) = q, \\ \& P(S_{n+1} = j+1 \mid S_n = j) &= P(X_{n+1} = 1) = p.\end{aligned}$$

$\therefore$ The $P(S_{n+1})$ is only dependent on $P(S_n)$ & not of $S_{n+1}, S_{n-2}, \dots, S_1$.

$\therefore$ The above example is a Markov

process.

Markov Chain:

Let $\{X_n \mid n=0,1,2,\dots\}$ be a sequence of r.v. Let the possible outcome of X_n be j , $j=0,1,2,\dots$.

The stochastic process $\{X_n \mid n=0,1,2,\dots\}$ is called a Markov chain, if for $j, k, j_1, \dots, j_{n-1} \in \mathbb{N}$.

$$\begin{aligned} P(X_n = k \mid X_{n-1} = j, X_{n-2} = j_1, \dots, X_0 = j_m) \\ = P(X_n = k \mid X_{n-1} = j) = p_{jk}. \end{aligned}$$

$\downarrow$ Future $\downarrow$ present

$\underbrace{\hspace{10em}}_{\text{past}}$

The outcomes X_n are called the state of Markov chain.

If X_n has outcome j , i.e. $X_n = j$, the process is said to be at state j .

at n^{th} trial.

The probability p_{jk} from state j at n^{th} trial to the state k at $(n+1)^{\text{th}}$ trial is called transition probabilities.

If the transition probabilities are independent on n , the Markov chain is called homogeneous or stationary transition probabilities.

If they depend on n , its called non-homogeneous.

Transition Matrix:

The transition probabilities satisfy,

$$p_{jk} \geq 0 \quad \& \quad \sum_k p_{jk} = 1 \quad \text{for all } j.$$

The matrix

$$P = \begin{bmatrix} p_{11} & p_{12} & \dots & p_{1k} & \dots \\ p_{21} & p_{22} & \dots & p_{2k} & \dots \\ \vdots & \vdots & & \vdots & \\ \vdots & \vdots & & \vdots & \end{bmatrix} \text{ is}$$

called transition probability matrix of the Markov chain $\{X_n \mid n \in \mathbb{N}\}$.

Ex: A particle performs a random walk with absorbing barriers, say, at 0 & 4. Whenever it is at any position r , $0 < r < 4$, it moves to $r+1$ with probability p or to $r-1$ with probability q , $p+q=1$. But as soon as it reaches 0 or 4 it remains there itself. Let X_n be position of particle after n moves. Determine the transition matrix.

Sol: We first write transition probabilities. For $0 < r < 4$

$$P \{ X_n = r+1 \mid X_{n-1} = r \} = p$$

$$P \{ X_n = r-1 \mid X_{n-1} = r \} = q$$

$$P \{ X_n = r \mid X_{n-1} = r \} = 0$$

For $r = 0, 4$

$$P \{ X_n = 0 \mid X_{n-1} = 0 \} = 1$$

$$P \{ X_n = 4 \mid X_{n-1} = 4 \} = 1$$

Transition matrix is

		States of X_n					
		0	1	2	3	4	
States of X_{n-1}	0	1	0	0	0	0	
	1	q	0	p	0	0	
	2	0	q	0	p	0	
	3	0	0	q	0	p	
	4	0	0	0	0	1.	

2. Suppose that coin with probability p for head is tossed indefinitely. Let X_n be the outcome of the n th trial, be k , where k denotes that there is a run of k successes (ie length of the uninterrupted block of heads is k).

$$p_{jk} = P(X_n = k \mid X_{n-1} = j) = \begin{cases} p, & k = j+1 \\ q, & k = 0 \\ 0, & \text{otherwise} \end{cases}$$

Write the transition matrix.

Sol:

		States of X_n						
		0	1	2	3	...	k	...
States of X_{n-1}	0	q	p	0	0	...	0	...
	1	q	0	p	0	...	0	...
	2	q	0	0	p	...		
	3	q	0	0	0			
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$			
	k	q	$\vdots$	$\vdots$	$\vdots$			
	$\vdots$							

A box contains 3 silver coins and 3 gold coins. The following experiment is repeated indefinitely.

A coin is selected at random from the box. If coin is silver, it is put back into box. If it is gold, it is not replaced.

Let X_n denote the no of gold coins remaining in the box after n -selections.

Find the transition probability ^{matrix} of the Markov chain $\{X_n\}$

Sol: Since X_n is no of gold coins remaining in box & there are 3 gold coins, the outcomes of each X_n is $\{0, 1, 2, 3\}$

States of X_{n+1} represents no of gold remaining after $n+1$ selections [current situation].

X_n represents no of gold going to remain in next selection.

		States of X_n			
		0	1	2	3
States of X_{n-1}	0				
	1				
	2	0	$\frac{2}{5}$	$\frac{3}{5}$	0
	3	0	0	$\frac{1}{2}$	$\frac{1}{2}$

$P(X_n = 3 \mid X_{n-1} = 3)$ is same as probability of choosing a silver coin from 6 coins $= \frac{3}{6} = \frac{1}{2}$

$$\therefore P(X_n = 3 \mid X_{n-1} = 3) = \frac{1}{2}$$

$P(X_n = 2 \mid X_{n-1} = 3)$ is same as probability of choosing a gold coin from 6 coins $= \frac{3}{6} = \frac{1}{2}$.

$$P(X_n = 1 \mid X_{n-1} = 3) = 0 = P(X_n = 0 \mid X_{n-1} = 3)$$

$$P(X_n = 3 \mid X_{n-1} = 2) = 0$$

$P(X_n = 2 \mid X_{n-1} = 2)$ is same as
probability of choosing silver coin
from 5 coins $= \frac{3}{5}$

$$P(X_n = 1 \mid X_{n-1} = 2)$$

A basket contains 4 apples & 2 oranges. The following experiment is repeated indefinitely:

A fruit is chosen at random from basket. If fruit is apple it is put back. If it is an orange, it is not replaced.

Let X_n denote the no of oranges remaining in the basket after n selections. Find the transition probability matrix of the Markov chain $\{X_n\}$.

Solution: Since X_n is no of oranges remaining, and no of oranges are 2 the outcome of X_n is 0, 1, 2.

		State space of X_n		
		0	1	2
State space of X_{n-1}	0	1	0	0
	1	$\frac{1}{5}$	$\frac{4}{5}$	0
	2	0	$\frac{2}{6}$	$\frac{4}{6}$

Probability distribution:

The probability of $X_r, X_{r+1}, \dots, X_{r+n}$ can be computed in terms of transition probabilities and the initial probability X_r .

$$\begin{aligned} & P(\{X_r = a, X_{r+1} = b, \dots, X_{n-2} = i, X_{n-1} = j, X_n = k\}) \\ &= P(\{X_n = k \mid X_{n-1} = j, \dots, X_r = a\}) \cdot P(\{X_r = a, \dots, X_{n-1} = j\}) \\ &= P(\{X_n = k \mid X_{n-1} = j\}) \cdot P(\{X_{n-1} = j \mid X_{n-2} = i, \dots, X_r = a\}) P(X_r = a, \dots, X_{n-2} = i) \\ &= P(X_n = k \mid X_{n-1} = j) \cdot P(X_{n-1} = j \mid X_{n-2} = i) \cdot \dots \cdot P(X_r = a) \\ &= P(X_r = a) \cdot P_{jk} P_{ij} \dots \end{aligned}$$

$$P(X_n, X_{n-1}, \dots, X_r) = P(X_n \mid X_{n-1}) \cdot P(X_{n-1}, \dots, X_r)$$

Question: Let $\{X_n \mid n \geq 0\}$ be Markov chain with 3 states 0, 1, 2 & with transition matrix

$$\begin{bmatrix} 3/4 & 1/4 & 0 \\ 1/4 & 1/2 & 1/4 \\ 0 & 3/4 & 1/4 \end{bmatrix}$$

and initial probability distribution $P(X_0 = i) = \frac{1}{3} \quad i = 0, 1, 2.$

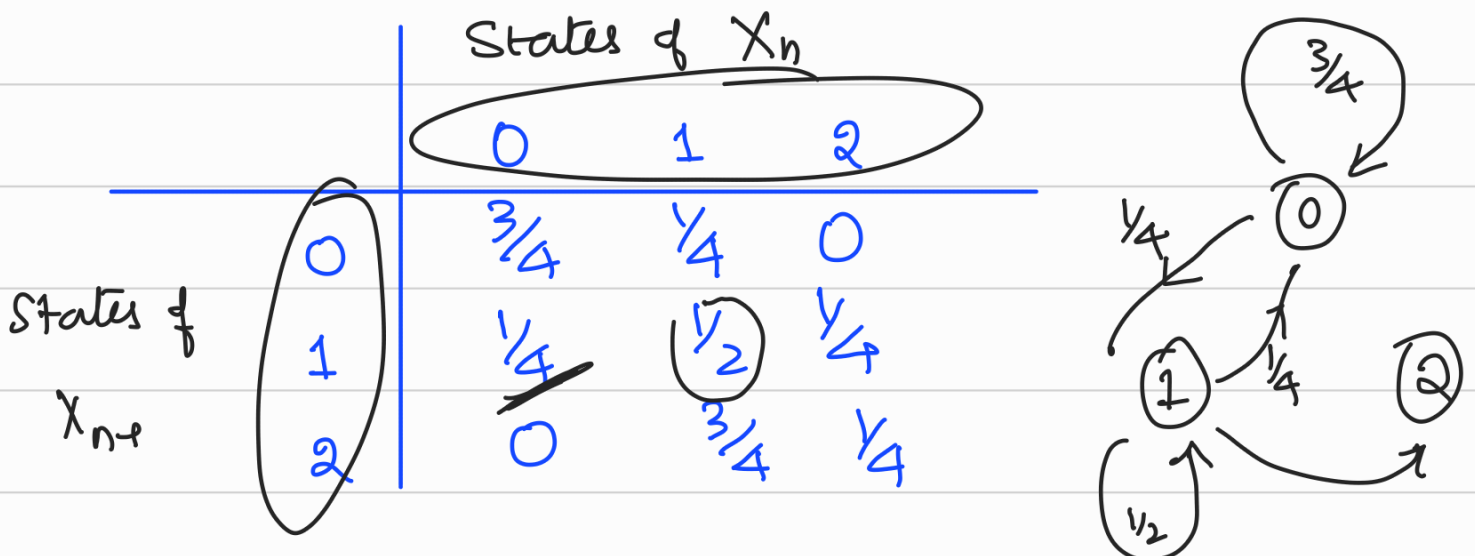
Evaluate: i) $P(X_1 = 1 \mid X_0 = 2)$

ii) $P(X_2 = 2, X_1 = 1, X_0 = 2)$

iii) $P(X_3 = 1, X_2 = 2, X_1 = 1, X_0 = 2)$

iv) $P(X_2 = 2, X_1 = 1 \mid X_0 = 2)$

Sol: Given transition matrix is



$$i) P(X_1=1 | X_0=2) = 3/4$$

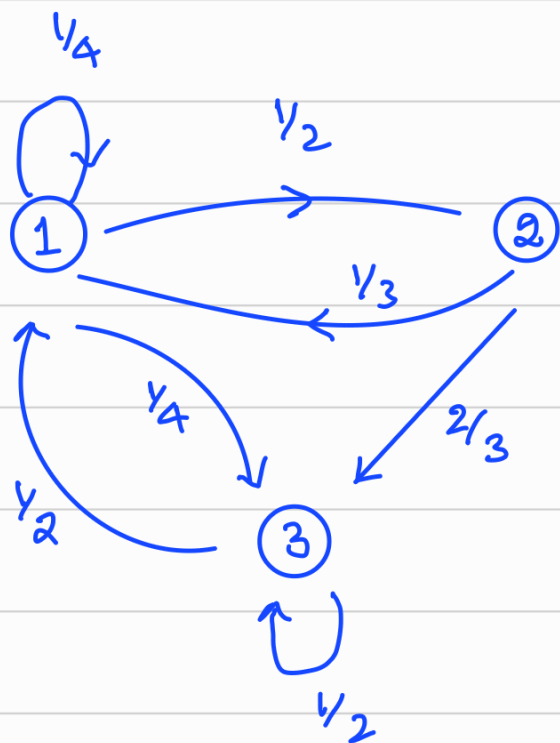
$$\begin{aligned} ii) P(X_2=2, X_1=1, X_0=2) \\ P(X_2=2 | X_1=1, X_0=2) \cdot P(X_1=1, X_0=2) \\ = P(X_2=2 | X_1=1) \cdot P(X_1=1 | X_0=2) \cdot P(X_0=2) \\ = \frac{1}{4} \times \frac{3}{4} = \frac{1}{3} = \frac{1}{16} \end{aligned}$$

$$\begin{aligned} iii) P(X_3=1, X_2=2, X_1=1, X_0=2) \\ = P(X_3=1 | X_2=2) \cdot P(X_2=2, X_1=1, X_0=2) \\ = P(X_3=1 | X_2=2) \cdot P(X_2=2 | X_1=1) \\ \cdot P(X_1=1, X_0=2) \\ = P(X_3=1 | X_2=2) \cdot P(X_2=2 | X_1=1) \cdot P(X_1=1 | X_0=2) \cdot P(X_0=2) \\ = \frac{3}{4} \times \frac{1}{4} \times \frac{3}{4} \times \frac{1}{3} = \frac{3}{64} \end{aligned}$$

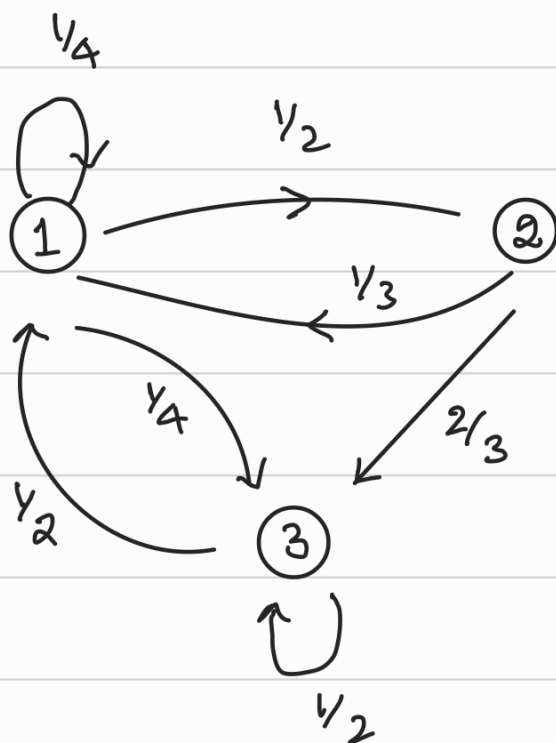
$$\begin{aligned} iv) P(X_2=2, X_1=1 | X_0=2) \\ = \frac{P(X_2=2, X_1=1, X_0=2)}{P(X_0=2)} = \frac{\frac{1}{16}}{\frac{1}{3}} = \frac{3}{16} \end{aligned}$$

Markov Chains as Graphs:

1. The state of a Markov chain is represented by vertex of a graph.
2. The directed edge from vertex i to j is represented by arrow $i \rightarrow j$ with weight p_{ij}



Q. The transition diagram is as follows:



If we know $P(X_0=1) = \frac{1}{3}$. Find

i) Transition probability matrix

ii) $P(X_1=1 | X_0=2)$

iii) $P(X_0=1, X_1=2, X_2=3)$.

iv) $P(X_2=3, X_1=2 | X_0=1)$

Solution. i) Transition probability matrix

is $P =$

$$\begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix} \end{matrix}$$

$$\text{iii) } P(X_0=1, X_1=2, X_2=3) \\ = P(X_2=3, X_1=2, X_0=1)$$

$$= P(X_2=3|X_1=2) \cdot P(X_1=2|X_0=1) \cdot P(X_0=1)$$

$$= \frac{2}{3} \cdot \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{9}$$

Poisson process:

A stochastic process $\{X(t) \mid t \geq 0\}$ is called a Poisson process with rate parameter $\lambda > 0$ if it satisfies following:

1. $N(0) = 0$ [No events has occurred at time zero]
2. The number of events occurring in disjoint time intervals are independent.

For example, the no of telephone calls received b/w 0-2 minutes & b/w 2-5 minutes are independent.

3. Poisson distribution counts:

The no of events in any interval of length t follows a Poisson distribution with mean length λt :

$$P[X(t) = k] = \frac{(\lambda t)^k e^{-\lambda t}}{k!} \quad k=0, 1, 2, \dots$$

where λ is mean rate [average no. of events per unit time]

λt = expected no of events in time t

$X(t)$ = total no of events up to time t .

Q. Customers arrive at a service counter according to a Poisson process with an average rate of 5 customers per hour. Find probability that in a half-hour (30 minutes) period,

i) exactly 2 customers arrive

ii) at least 3 customers arrive.

Sol: Given average rate

$\lambda = 5$ customers per hour.

Expected no of customers in half hour is $\lambda t = 5 \times 0.5 = 2.5$

$$i) P[X(0.5) = 2] = \frac{(2.5)^2 \cdot e^{-2.5}}{2!} = 0.2565$$

$$ii) P[X(0.5) \geq 3] = 1 - P[X(0.5) \leq 2]$$

$$= 1 - [P(X=0) + P(X=1) + P(X=2)]$$

$$= 1 - \left[\frac{(2.5)^0 \cdot e^{-2.5}}{0!} + \frac{(2.5)^1 \cdot e^{-2.5}}{1} + \frac{(2.5)^2 \cdot e^{-2.5}}{2!} \right]$$

$$= 1 - e^{-2.5} \left[1 + \frac{2.5}{1} + \frac{(2.5)^2}{2!} \right] = 0.457$$

Q. A machine experiences failures according to a Poisson process at an avg rate of 1.2 failures per day. Find the probability that during a 5-day period.

i) the machine fails no more than 2-times.

ii) the machine fails exactly 2-times

iii) the machine fails exactly 6-times.

Sol: Average is $\lambda = 1.2$ failures / day.
time period is $t = 5$ days.

$$\therefore \lambda t = 1.2 \times 5 = 6$$

$$i) P(X(5) \leq 2) = P(N(5)=0) + P(N(5)=1) + P(N(5)=2)$$

$$= \frac{(6)^0 e^{-6}}{0!} + \frac{(6)^1 e^{-6}}{1!} + \frac{6^2 \cdot e^{-6}}{2!}$$

$$= \underline{\underline{0.06197}}$$

$$ii) P(X=2) = \frac{(6)^2 \cdot e^{-6}}{2!} = \underline{\underline{0.0446}}$$

$$iii) P[X(5)=6] = \frac{6^6 \cdot e^{-6}}{6!} = 0.1606$$